



A Collection of Counting Theorems and Lemmas

Drew Davidson, Calvin Mar, Ben Wafford

Olin, Department of Mathematics
Centre College
Danville, KY

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OUTLINE

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Combinatorics

Stars and Bars Counting Method

Theorem: The number of ways to place n indistinguishable balls into k distinguishable boxes is:

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

We represent the n balls by n stars and insert $k - 1$ bars in between the stars to separate them into k groups, i.e., if we had 7 balls and 4 boxes they could be represented as such:

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There are a total of $n + k - 1$ positions with n stars and $k - 1$ bars. Thus, the number of ways to place n balls into k boxes is the same as the number of ways to choose n positions among $n + k - 1$ spaces for the stars with all the remaining positions taken as bars. [Mei Li,]

Lemma 4.1

For $0 < |r| < 1$,

$$\frac{1}{(1-r)^p} = \sum_{j=0}^{\infty} \frac{(p+j-1)!}{j!(p-1)!} r^j.$$

- This Lemma is particularly useful when counting linearly independent homogeneous polynomials. The proof of this Lemma will help us to better understand this.
- This proof will use both the Stars and Bars counting method as well as Cauchy Products.
- Recall Cauchy Products of Power Series:

$$\left(\sum_{i=0}^{\infty} a_i x^i\right) \cdot \left(\sum_{j=0}^{\infty} a_j x^j\right) = \sum_{k=0}^{\infty} c_k x^k$$

[Frye and Efthimiou, 2012]

Proposition 4.2

If $K(p, n)$ denotes the number of linearly independent homogeneous polynomials of degree n in p variables, then

$$K(p, n) = \frac{(n + p - 1)!}{n!(p - 1)!}$$

A homogeneous polynomial of degree n is a sum of terms in which the exponents of the variables of each term sum to n .

This basically means:

"How many ways can we place n indistinguishable rocks into p distinguishable bins?"[Frye and Efthimiou, 2012]

Lemma 4.3

Lemma 4.3 states that

$$\binom{k}{l} = \frac{k}{l} \binom{k-1}{l-1}$$

In order to prove this expand the right side of the equation and compute:

$$\begin{aligned} \frac{k}{l} \binom{k-1}{l-1} &= \frac{k}{l} \cdot \frac{(k-1)!}{(l-1)!(k-l)!} \\ &= \frac{k \cdot (k-1)!}{l \cdot (l-1)!(k-l)!} = \frac{k!}{l!(k-l)!} = \binom{k}{l} \\ \therefore \binom{k}{l} &= \frac{k}{l} \binom{k-1}{l-1} \end{aligned}$$

[Frye and Efthimiou, 2012]

Theorem 4.4

Linearly Independent Homogeneous Harmonic Polynomials

Theorem: If $N(p, n)$ denotes the number of linearly independent homogeneous harmonic polynomials of degree n in p variables, then

$$N(p, n) = \frac{2n + p - 2}{n} \binom{n + p - 3}{n - 1}$$

The theorem provides an equation to determine the number of linearly independent homogeneous harmonic polynomials given specific values that are possible. This is the number of polynomials that are needed to form a basis.

[Frye and Efthimiou, 2012]

Theorem 4.4

Proof

Let H_n be a homogeneous harmonic polynomial of degree n in p variables

$$H_n(x_1, x_2, \dots, x_p) = \sum_{j=0}^n x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1})$$

Using the Laplace operator and since H_n is harmonic,

$$\begin{aligned} 0 = \Delta_p H_n &= \left(\frac{\delta^2}{\delta x_p^2} + \Delta_{p-1} \right) H_n = \sum_{j=2}^n j(j-1) x_p^{j-2} h_{n-j} + \sum_{j=0}^n \Delta_{p-1} (x_p^j h_{n-j}) \\ &= \sum_{j=0}^n x_p^j ((j+2)(j+1) h_{n-j-2} + \Delta_{p-1} h_{n-j}) \end{aligned}$$

Theorem 4.4

Proof Conclusion

Since each x^j is linearly independent we can create a system of linear equations that can be solved by deciding h_n and h_{n-1} .

Using Proposition 4.2 to count the number of linear combination for h_n and h_{n-1} allows for the following

$$N(p, n) = K(p-1, n) + K(p-1, n-1) = \binom{p+n-2}{2} + \binom{p+n-3}{n-1}$$

By Lemma 4.3, we can rewrite this as

$$N(p, n) = \frac{2n+p-2}{n} \binom{n+p-3}{n-1}$$

Concluding the proof

REFERENCES



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